POOJAA V K **B020**

Detecting Financial Shocks & Segmenting Behavioral Risk Profiles using Unsupervised Learning

UNSUPERVISED MACHINE LEARNING PROJECT

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1. Abstract

This project presents a complete unsupervised machine learning pipeline for the detection of financial shocks and the construction of interpretable behavioural risk profiles from unlabelled mobile money transaction data. Using a 100,000-transaction subset of the PaySim synthetic dataset, calibrated on real M-Pesa transaction logs, the pipeline proceeds through four analytically distinct stages: feature engineering, anomaly detection, clustering-based segmentation, and composite risk scoring.

Fifteen transaction-level features are engineered from raw balance dynamics, capturing full account drain events, balance change ratios, destination credit patterns, and contextual deviations from type-specific and time-step-specific norms. Financial shocks are detected using an ensemble of Isolation Forest and DBSCAN, which together flag 5,042 transactions (5.04% of the dataset) as anomalous. The combined detector leverages complementary detection mechanisms—global isolation and local density—resulting in improved anomaly coverage compared to either model in isolation.

K-Means clustering with K = 4 (selected via Silhouette Score optimisation) segments the transactions into four behavioural personas: Stable, Moderate Risk, Shock-Prone, and High Risk. These segments exhibit clearly differentiated risk characteristics, with shock rates of 0.10%, 1.27%, 99.84%, and 100%, respectively.

A composite Financial Stress Index (FSI), constructed as a weighted linear combination of shock status, balance drain ratio, and log-transformed transaction amount, provides a continuous and interpretable risk signal. The FSI is validated through its monotonic relationship with key empirical risk indicators, including fraud rate, full drain frequency, and shock magnitude across defined risk tiers.

Overall, the project demonstrates that meaningful and operationally actionable risk segmentation can be achieved entirely without reliance on ground-truth fraud annotations, highlighting the effectiveness of unsupervised learning for financial risk discovery.

2. Introduction

The rapid growth of digital payment systems in emerging economies has transformed financial risk dynamics. Mobile money platforms such as M-Pesa process high volumes of transactions with near-instant settlement, enabling financial inclusion but also creating opportunities for fast, large-scale fraudulent activities that can bypass traditional rule-based systems.

Financial fraud and systemic shocks are inherently difficult to detect because they are embedded within large volumes of legitimate transactions and cannot be identified using any single feature. A transaction becomes anomalous only when multiple signals co-occur—such as high value, complete balance depletion, absence of destination credit, and deviation from expected behavioural patterns—forming statistically unlikely combinations.

To address this, behavioural risk profiling shifts the focus from binary fraud detection to segment-level analysis. Instead of classifying transactions as simply fraudulent or not, it groups them into risk-based behavioural segments, enabling more nuanced monitoring and proactive intervention.

Unsupervised learning is particularly suited to this problem because true shock labels are unavailable in real-world systems, and fraud patterns evolve continuously. Unlike supervised models, unsupervised methods can adapt to emerging risk structures without relying on predefined labels.

Accordingly, this project implements a four-stage pipeline: (1) feature engineering to capture transactional behaviour, (2) anomaly detection using Isolation Forest and DBSCAN to identify shock events, (3) clustering with K-Means and Gaussian Mixture Models to derive behavioural risk segments, and (4) construction of a Financial Stress Index to produce a continuous and interpretable risk score for each transaction.

3. Problem Statement

The objective of this project is to detect financial shocks and segment transactions into interpretable behavioural risk profiles using a fully unsupervised approach, without relying on fraud labels.

A financial shock is defined as a transaction that significantly deviates from expected behavioural patterns across multiple dimensions, such as unusually large amounts, near-complete depletion of the sender's balance, lack of corresponding credit to the receiver, and deviation from type- and time-specific norms. While such events may indicate fraud, the focus is on identifying statistical anomalies and their associated risk.

This problem is challenging due to extreme class imbalance (0.141% fraud rate), absence of repeated user activity for most accounts, and weak individual feature correlation with fraud, implying that risk is embedded in complex feature interactions rather than single variables.

Therefore, the problem is framed as developing a pipeline that, given unlabelled transaction data,

(a) detects anomalous shock events,

(b) segments transactions into risk-ranked behavioural groups,

(c) produces a continuous composite risk score, all derived purely from transaction structure without using fraud labels.

4. Objectives

The project is guided by the following key objectives:

- To engineer transaction-level features capturing multi-dimensional characteristics of financial shocks, including balance dynamics, amount-to-balance ratios, structural anomaly indicators, and contextual deviations.
- To apply Isolation Forest on the full dataset to generate a continuous anomaly score and corresponding binary shock label for each transaction.
- To apply DBSCAN on a representative sample for density-based anomaly detection and combine its outlier labels with Isolation Forest results to form a consolidated shock indicator.
- To assess agreement and complementarity between the two anomaly detection methods using the Jaccard similarity index.
- To determine the optimal number of clusters for K-Means using the Silhouette Score, ensuring a minimum of four clusters for meaningful segmentation.
- To perform clustering using K-Means and Gaussian Mixture Models on an enriched feature set, and select the better model based on Silhouette Score and Davies–Bouldin Index.
- To validate cluster structures through dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-SNE.
- To assign interpretable behavioural personas—Stable, Moderate Risk, Shock-Prone, and High Risk—based on cluster-level risk characteristics.
- To construct a Financial Stress Index (FSI) as a composite risk metric and validate its alignment with empirical risk indicators.
- To generate a final enriched dataset containing engineered features, anomaly labels, cluster assignments, persona classifications, and FSI scores for downstream risk monitoring applications.

5. Dataset Description

5.1 Overview of the PaySim Dataset

PaySim is a synthetic mobile money transaction simulator developed by Lopez-Rojas, Elmir, and Axelsson (2016), calibrated using real transaction data from M-Pesa, one of the largest mobile money platforms operating in Kenya and Tanzania. It generates statistically realistic transaction streams that preserve distributional properties, temporal behaviour, and fraud patterns of real-world data while ensuring complete anonymity.

5.2 Dataset Subset

This project uses a random sample of 100,000 transactions from the full PaySim dataset of approximately 6.3 million records. The subset ensures computational efficiency, particularly for algorithms such as DBSCAN, while retaining sufficient variability for meaningful analysis. The dataset spans 736 simulation steps (approximately 30 days, with each step representing one hour) and includes all five transaction types. The fraud rate of 0.141% closely matches the full dataset (~0.13%), confirming that the subset is representative.

5.3 Feature Descriptions

Feature	Type	Description
step	Integer	Simulation timestep (1–736), representing hours; used for temporal analysis
type	Categorical	Transaction type: CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER
amount	Float	Transaction amount in local currency
nameOrig	String	Anonymised sender account ID
oldbalanceOrg	Float	Sender balance before transaction
newbalanceOrig	Float	Sender balance after transaction
nameDest	String	Anonymised receiver account ID
oldbalanceDest	Float	Receiver balance before transaction
newbalanceDest	Float	Receiver balance after transaction
isFraud	Binary	Ground-truth fraud label (used only for validation)
isFlaggedFraud	Binary	System flag for transactions exceeding 200,000 units

Table 1: PaySim Raw Feature Descriptions

6. Methodology

6.1 Data Preprocessing

The PaySim dataset requires minimal preprocessing due to its synthetic nature. The 100,000-transaction sample contains no missing values or duplicates, eliminating the need for imputation or deduplication.

The transaction type variable is encoded into numerical form using categorical encoding, creating a type_encoded feature suitable for model input. The dataset is sorted chronologically by the step variable and reindexed to maintain temporal consistency for sequential analysis.

Feature scaling is applied at multiple stages. The 11-feature matrix used for anomaly detection is standardised before applying Isolation Forest and DBSCAN. Similarly, the 15-feature matrix used for clustering is independently standardised prior to K-Means and Gaussian Mixture Model training. Additionally, the components of the Financial Stress Index (FSI) are normalised to a [0, 1] range before aggregation. This ensures balanced feature contribution across all models.

6.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand data distributions, identify structural patterns, and guide feature engineering. Analysis of transaction types shows that CASH_OUT (approximately 35,602 transactions) and PAYMENT dominate the dataset in terms of volume. Although TRANSFER transactions occur less frequently, they exhibit a disproportionately higher fraud rate, making transaction type a critical feature for anomaly detection. These observations provide important insights into transactional behaviour and inform the design of features used in subsequent modelling stages.

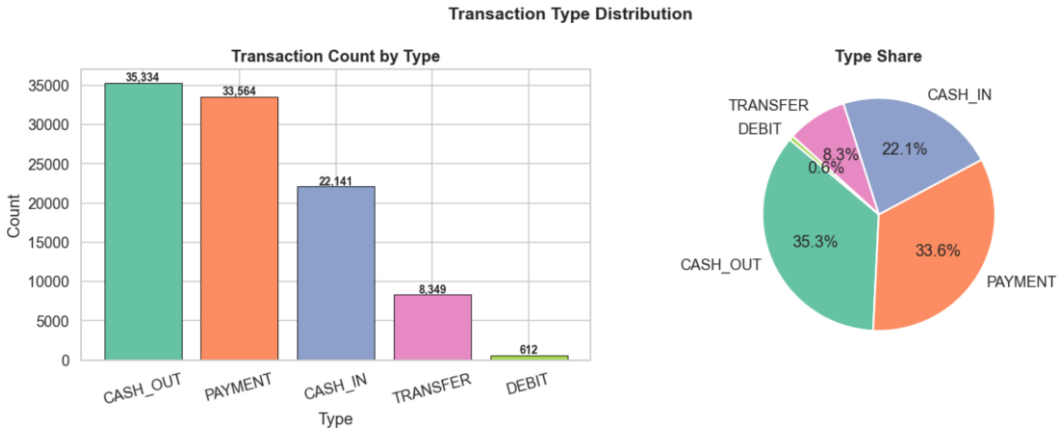

Figure 1: Transaction Count by Type and Fraud Rate by Type

The distribution of transaction amounts is highly right-skewed, with the majority of transactions concentrated at lower values and a long tail of large transactions. Applying a log(1 + x) transformation produces an approximately normal distribution, supporting the use of log_amount as a stable and effective continuous feature for modelling. Temporal analysis of transaction volume alongside fraud occurrence shows that fraudulent transactions are distributed throughout the entire simulation period rather than being concentrated in high-activity intervals. This indicates that transaction volume alone is insufficient as a detection signal and reinforces the need for multi-dimensional anomaly detection.

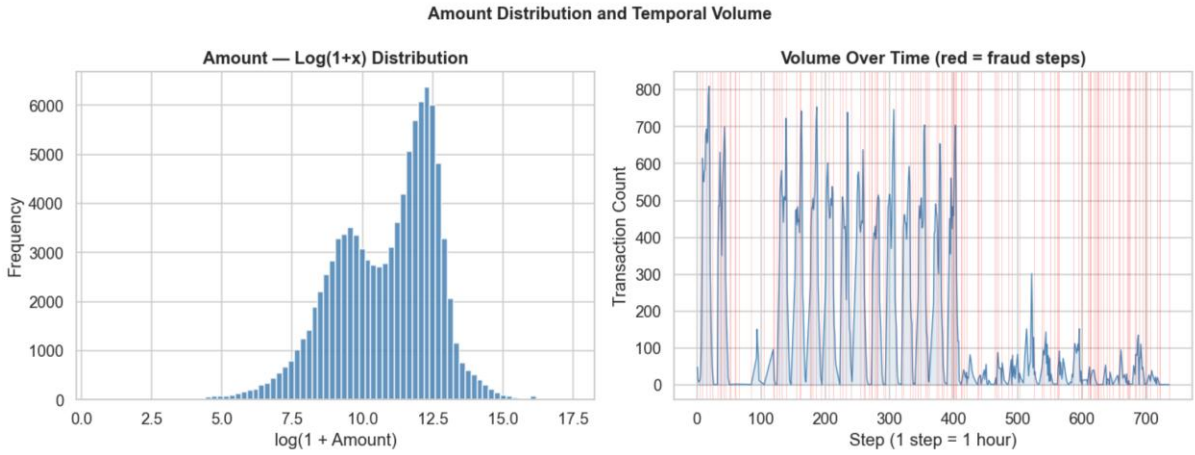

Figure 2: Log-Transformed Amount Distribution and Temporal Transaction Volume with Fraud Steps

Correlation analysis of the eight numerical raw features shows that isFraud has only moderate pairwise correlation with individual variables, with the strongest relationships observed for newbalanceOrig and related balance features. This confirms that no single feature is sufficient to capture fraudulent behaviour. These findings support the need for multi-dimensional feature engineering, where risk signals are derived from combinations of variables rather than isolated attributes.

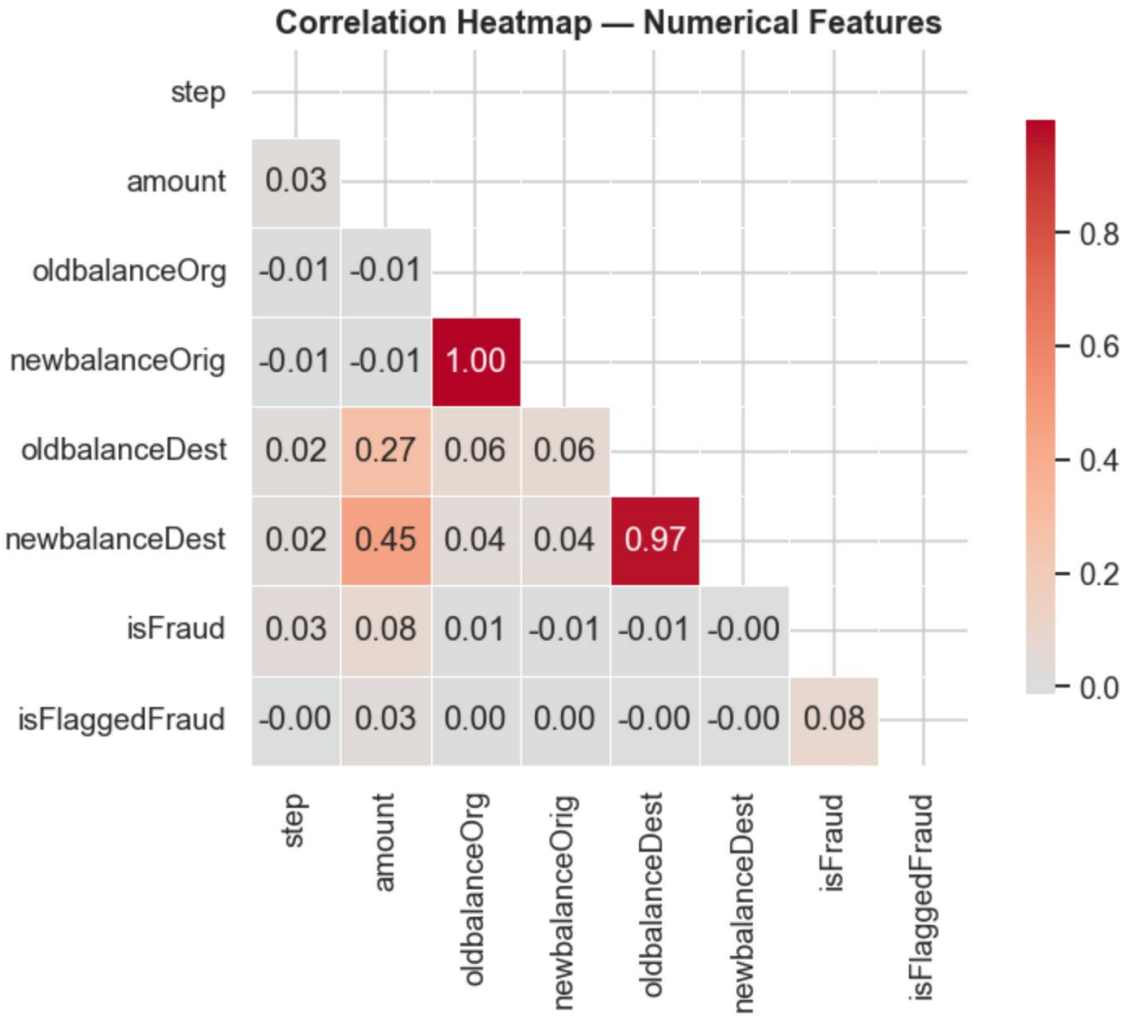

Figure 3: Correlation Heatmap of Numerical Raw Features

6.3 Feature Engineering

Since most accounts appear only once in the dataset, user-level historical features cannot be computed. Therefore, feature engineering focuses on transaction-level signals that capture structural characteristics of financial anomalies.

Balance Dynamics

- balance_drain: Absolute reduction in originator balance (oldbalanceOrg - newbalanceOrig)
- balance_change_ratio: Ratio of balance drain to pre-transaction balance
- surplus_after: Ratio of post-transaction balance to pre-transaction balance
- balance_delta_dest: Change in destination balance (newbalanceDest - oldbalanceDest)
- dest_balance_ratio: Ratio of destination balance change to pre-transaction destination balance

Amount-to-Balance Ratio

- amount_to_orig_bal: Ratio of transaction amount to originator balance
- log_amount: Log(1 + x) transformation of transaction amount

Binary Shock Indicators

- full_drain: 1 if balance_change_ratio > 0.999
- no_dest_credit: 1 if balance_delta_dest ≤ 0
- amount_mismatch: 1 if amount deviates from balance delta beyond 1% tolerance

Contextual Deviation Features

- step_tx_count: Number of transactions in the same timestep
- step_mean_amount: Mean transaction amount in the timestep
- deviation_from_type_mean: Deviation from mean amount for that transaction type

Temporal Features

- hour: Step modulo 24
- day: Step divided by 24

The binary indicators (full_drain, no_dest_credit, amount_mismatch) capture key structural anomaly signals and contribute strongly to shock detection. All features are validated to ensure no null or infinite values before model training.

6.4 Models Used

6.4.1 Isolation Forest

Isolation Forest is an anomaly detection algorithm that isolates observations using random feature splits. Anomalies require fewer splits and therefore have shorter path lengths. It is applied to the full dataset using the 11-feature matrix. The contamination parameter is set to 0.05, resulting in 5% of transactions being flagged as anomalies. The model uses 100 trees and a fixed random state.

6.4.2 DBSCAN

DBSCAN is a density-based algorithm that identifies clusters as dense regions and labels sparse points as outliers.

It uses two parameters:

- eps (neighbourhood radius) = 1.2
- min_samples = 10

Due to its quadratic complexity, DBSCAN is applied to a random sample of 8,000 transactions. Points labelled as noise (−1) are treated as anomalies. Non-sampled points are assigned a default non-anomalous label.

6.4.3 K-Means Clustering

K-Means partitions data into K clusters by minimising within-cluster variance. It is applied on the full dataset using the 15-feature matrix. The optimal number of clusters is selected using the Silhouette Score over K = 2 to 9, with a minimum constraint of 4, resulting in K = 4. The model uses 10 initialisations and a maximum of 300 iterations.

6.4.4 Gaussian Mixture Model

The Gaussian Mixture Model represents data as a mixture of multiple Gaussian distributions and uses the Expectation-Maximisation algorithm for parameter estimation. Unlike K-Means, it supports non-spherical clusters and provides probabilistic cluster assignments. The model is applied with K = 4, full covariance matrices, 3 initialisations, and up to 200 iterations. Final labels are assigned based on maximum posterior probability. Both K-Means and GMM are evaluated using Silhouette Score and Davies-Bouldin Index to determine the better clustering approach.

7. Model Implementation

7.1 Anomaly Detection Implementation

The anomaly detection stage is implemented as a two-model sequential pipeline, where both models operate on a standardised 11-feature matrix (X_anom). This matrix is constructed using StandardScaler on the selected anomaly features: log_amount, balance_change_ratio, surplus_after, amount_to_orig_bal, balance_delta_dest, full_drain, no_dest_credit, amount_mismatch, step_tx_count, type_encoded, and hour.

Isolation Forest is applied to the full dataset of 100,000 transactions with contamination set to 0.05 and 100 estimator trees. The model produces a continuous anomaly score (iso_score) using the decision function, along with a binary prediction where anomalous points are labelled as −1. These labels are converted into a binary indicator (iso_shock). In total, Isolation Forest identifies 5,000 transactions (5%) as anomalous, consistent with the contamination parameter.

DBSCAN is applied to a standardised subsample of 8,000 transactions using parameters eps = 1.2 and min_samples = 10. The model identifies low-density points as outliers, flagging 148 transactions in the sample as anomalies, representing approximately 0.15% of the total dataset.

A combined anomaly indicator (shock_event) is constructed by taking the union of iso_shock and dbscan_shock. This results in 5,042 detected shock events, of which 4,894 are identified only by Isolation Forest, 42 only by DBSCAN, and 106 by both models. The Jaccard similarity index between the two methods is 0.0210, indicating low overlap and strong complementarity, thereby improving overall anomaly coverage.

Finally, shock magnitude is computed as the absolute standardised deviation of the Isolation Forest score from its mean, providing a continuous measure of anomaly intensity. This metric is subsequently used in clustering and Financial Stress Index construction.

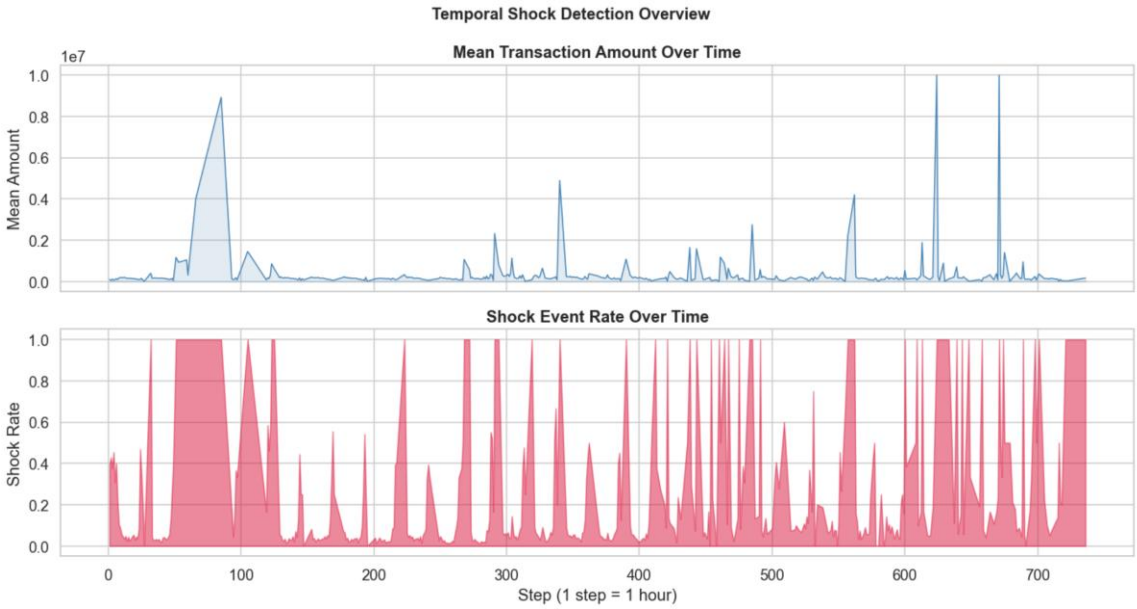

Figure 4: Temporal Distribution of Shock Events and Mean Transaction Amount over Simulation Steps

7.2 Clustering Implementation

The clustering feature matrix (X_cluster) extends the anomaly detection matrix by incorporating additional features: iso_score, shock_magnitude, step_mean_amount, and deviation_from_type_mean. This results in a 15-feature representation that captures both anomaly signals and broader transactional context. The matrix is independently standardised using StandardScaler to ensure balanced feature contribution.

The optimal number of clusters (K) is determined by evaluating K-Means over the range K = 2 to 9. Two criteria are used: inertia (for the elbow method) and Silhouette Score, computed on a 10,000-row sample to maintain computational efficiency. Both the elbow point in the inertia curve and the maximum Silhouette Score consistently indicate K = 4 as the optimal clustering configuration.

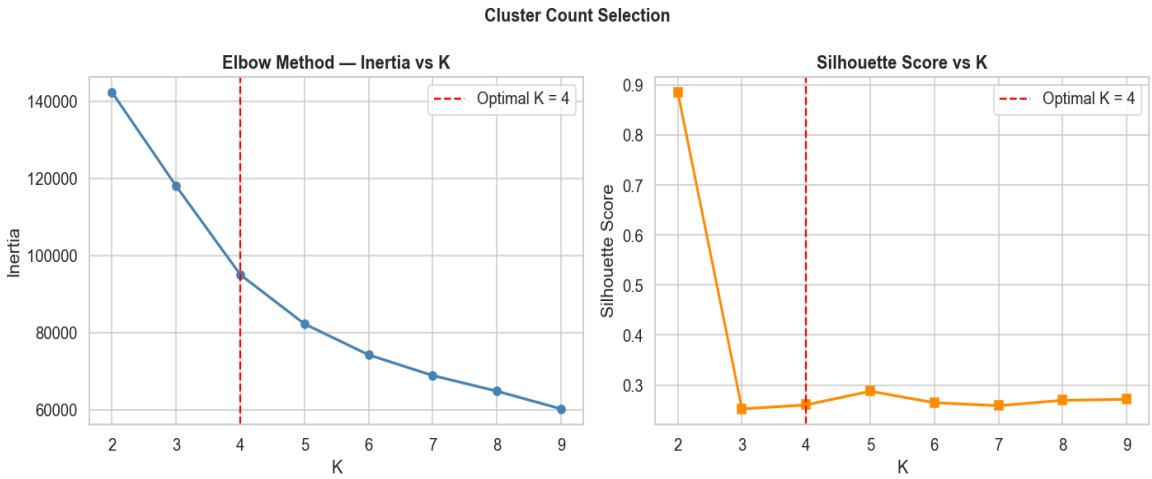

Figure 5: Elbow Method (Inertia vs K) and Silhouette Score vs K - Optimal K = 4

The final K-Means model is fitted with K = 4, n_init = 10, and max_iter = 300. The resulting cluster distribution is highly imbalanced, reflecting the natural structure of the dataset: Cluster 3 contains 60,358 transactions, Cluster 2 contains 35,097, Cluster 0 contains 4,432, and Cluster 1 contains 113.

A Gaussian Mixture Model (GMM) is also trained with K = 4 using full covariance matrices and three initialisations. Both models are evaluated on a 10,000-row sample using Silhouette Score and Davies–Bouldin Index. K-Means outperforms GMM on both metrics (Silhouette: 0.2822 vs 0.2733; DBI: 1.1851 vs 1.5393) and is therefore selected as the final clustering model. All subsequent analysis is based on K-Means cluster assignments.

7.3 Dimensionality Reduction and Visualisation

To analyse and validate cluster structure, two dimensionality reduction techniques are applied. Principal Component Analysis (PCA) is first used to project the 15-dimensional feature space into lower dimensions. Analysis shows that 9 principal components are required to explain 90% of the total variance, indicating that the dataset retains high intrinsic dimensionality.

t-Distributed Stochastic Neighbour Embedding (t-SNE) is applied to capture non-linear relationships and preserve local neighbourhood structure. Due to computational constraints, it is performed on a 3,000-row sample with perplexity set to 40, max_iter = 1000, and PCA-based initialisation.

Both PCA and t-SNE projections demonstrate clear separation between the four clusters. Overlaying anomaly labels shows that shock events are concentrated in a distinct region of the embedding space, providing strong visual validation of the clustering and anomaly detection pipeline.

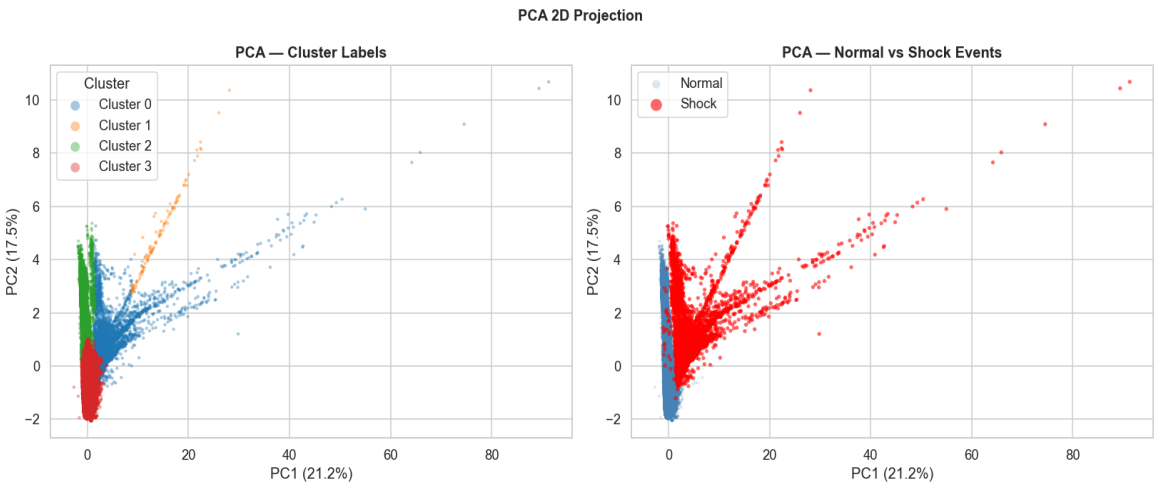

Figure 6: PCA 2D Projection Coloured by Cluster Label and Shock Event Status

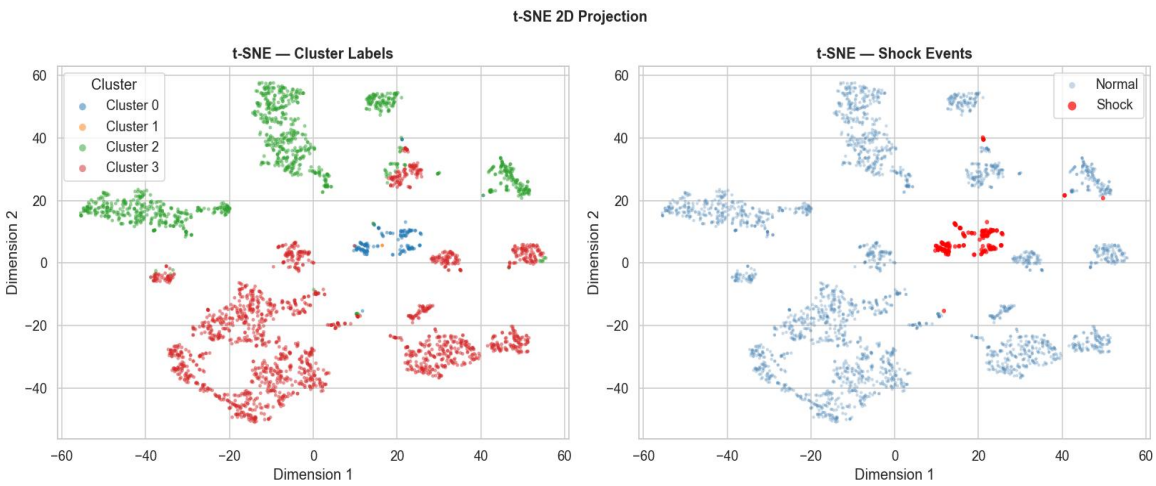

Figure 7: t-SNE 2D Projection Coloured by Cluster Label and Shock Event Status (3,000-transaction sample)

8. Evaluation Metrics

In the absence of ground-truth cluster labels, clustering performance is evaluated using internal validity metrics that assess the geometric structure of clusters. Two complementary measures are used: Silhouette Score and Davies-Bouldin Index.

8.1 Silhouette Score

The Silhouette Score measures how well each observation fits within its assigned cluster compared to neighbouring clusters. The final score is the mean across all observations. Values range from -1 to +1, where higher values indicate better clustering. For high-dimensional financial data, values between 0.2 and 0.4 are considered acceptable.

8.2 Davies-Bouldin Index

The Davies-Bouldin Index (DBI) measures cluster quality as the ratio of within-cluster scatter to between-cluster separation. For each cluster, it evaluates similarity with the most similar neighbouring cluster. Lower values indicate better clustering, as they reflect compact clusters with strong separation. Unlike the Silhouette Score, DBI penalises both high intra-cluster variance and poor inter-cluster separation simultaneously.

8.3 Results

Metric	**K-Means**	**GMM**	**Interpretation**
Silhouette Score (higher is better)	0.2822	0.2733	K-Means exhibits better cluster cohesion and separation
Davies-Bouldin Index (lower is better)	1.1851	1.5393	K-Means clusters are more compact and better separated
Selected Model	Yes	No	K-Means selected as the primary clustering model

Table 2: Clustering Evaluation Metrics - K-Means vs Gaussian Mixture Model

The superior performance of K-Means across both metrics indicates that the cluster structure in this dataset is relatively compact and well-separated. With the inclusion of anomaly-derived features, behavioural segments align more closely with centroid-based clustering assumptions than with the covariance-based flexibility of GMM.

The Silhouette Scores (approximately 0.27–0.28) are consistent with expectations for high-dimensional financial transaction data, where behavioural patterns overlap rather than form sharply distinct groups.

9. Results and Analysis

9.1 Cluster Characterisation and Persona Assignment

Clusters are assigned interpretable behavioural personas by ranking them using three key indicators: shock rate, full drain rate, and fraud rate. Based on this composite ranking, numeric cluster labels are replaced with meaningful persona names reflecting risk levels.

Cluster	Persona	N	Shock Rate	Avg Shock Magnitude	Full Drain Rate	Fraud Rate	Avg FSI
0	High Risk	4,432	99.84%	2.7755	33.60%	2.69%	0.9310
1	Stable	113	100.00%	5.0875	0.00%	0.00%	0.7812
2	Shock-Prone	35,097	1.27%	0.0249	13.61%	0.03%	0.4789
3	Moderate Risk	60,358	0.10%	0.0017	29.39%	0.02%	0.5164

Table 3: Cluster Profile Summary - Behavioral Personas and Key Risk Metrics

9.2 Behavioral Interpretation of Clusters

Cluster 3 (Moderate Risk, 60,358 transactions): This is the largest cluster (60.4% of all transactions). It shows a very low shock rate (0.10%) and fraud rate (0.02%), but a relatively high full drain rate (29.39%). This indicates that balance depletion is often legitimate in this segment, likely due to routine transactions such as payments from low-balance accounts. The average FSI (0.5164) reflects moderate risk without strong anomaly signals. This cluster represents normal transactional behaviour and requires standard monitoring.

Cluster 2 (Shock-Prone, 35,097 transactions): This cluster accounts for 35.1% of transactions and shows a low but notable shock rate (1.27%). The full drain rate (13.61%) and fraud rate (0.03%) remain low, while the shock magnitude is also minimal (0.0249). These transactions lie near the boundary of normal behaviour, indicating mild anomalies rather than severe risk. With an average FSI of 0.4789, this segment requires enhanced monitoring but not immediate action.

Cluster 0 (High Risk, 4,432 transactions): Almost all transactions (99.84%) are flagged as shocks in this cluster. It exhibits the highest average shock magnitude (2.7755), a significantly elevated fraud rate (2.69%), and a high full drain rate (33.60%). The average FSI (0.9310) is the highest across all clusters, indicating severe risk concentration. This cluster clearly represents high-risk activity and requires immediate investigation and intervention.

Cluster 1 (Stable, 113 transactions): This smallest cluster (0.1%) presents an unusual profile. All transactions are flagged as shocks with extremely high shock magnitude (5.0875), yet both fraud and full drain rates are zero. This suggests that these are structurally valid but statistically extreme transactions—possibly legitimate high-value transfers or institutional operations. Despite being labelled "Stable" due to zero fraud indicators, this cluster should still be reviewed due to its extreme anomaly scores.

9.3 Cluster Comparison Visualisation

Figure 8 illustrates the average values of four key indicators across clusters: log(amount), balance drain ratio, shock magnitude, and full drain rate. The visualisation highlights that the High Risk cluster dominates in anomaly intensity and structural risk, while the Stable cluster exhibits the highest transaction magnitudes. Moderate Risk and Shock-Prone clusters occupy intermediate behavioural positions, confirming the validity of the segmentation.

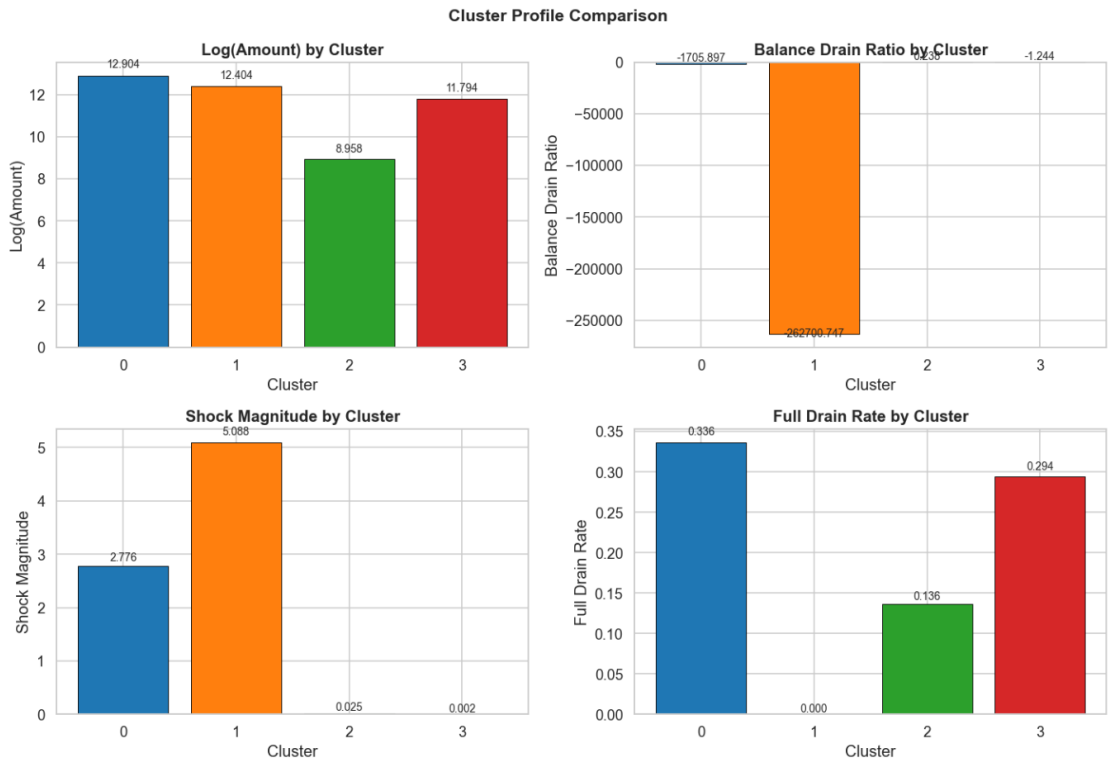

Figure 8: Mean Risk Indicators Across Clusters - Log Amount, Balance Drain Ratio, Shock Magnitude, Full Drain Rate

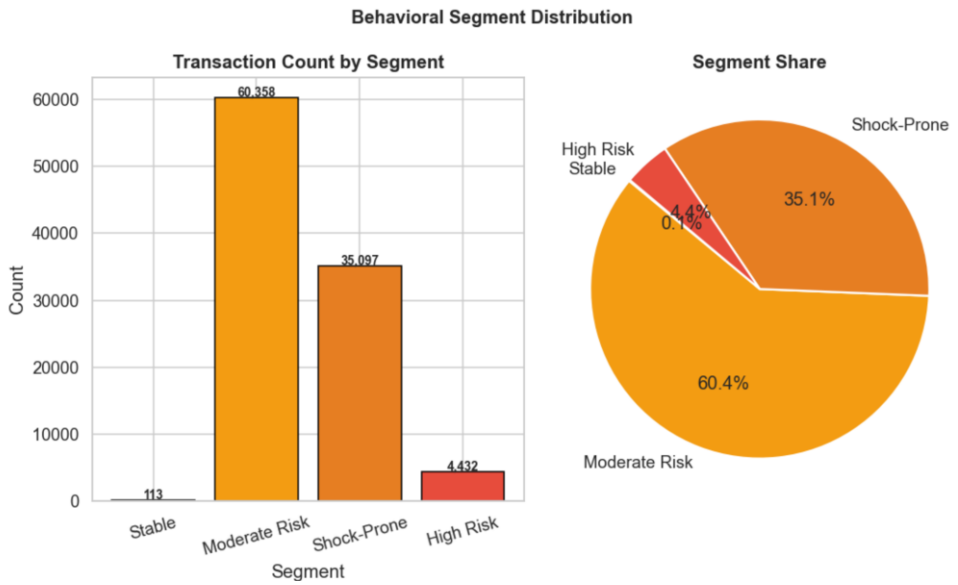

Figure 9: Segment Distribution by Transaction Count and Proportional Share

10.Financial Shock Detection Insight

10.1 Financial Stress Index Construction

The Financial Stress Index (FSI) is a composite, transaction-level risk score that combines three normalised indicators into a continuous value between 0 and 1. The weighting reflects the relative importance of each component. Shock status (0.40) carries the highest weight as it represents confirmed anomaly detection output. Balance change ratio (0.35) captures the proportional financial drain, a key structural indicator of risk. Log-transformed amount (0.25) provides a complementary measure of transaction magnitude. Each component is independently normalised to the range [0, 1] using MinMaxScaler before aggregation.

The resulting FSI has a mean of 0.5219 and a standard deviation of 0.0959, with values ranging from 0.35 to 1.00. The distribution exhibits a trimodal pattern, corresponding to three risk tiers defined using tercile thresholds.

Figure 10: FSI Score Distribution with Tier Boundaries, Tier Proportions, and FSI by Transaction Type

10.2 FSI Validation

The validity of the Financial Stress Index (FSI) is evaluated by analysing its relationship with key empirical risk indicators across defined risk tiers. A consistent monotonic increase in risk metrics across tiers confirms that the index captures meaningful financial risk.

FSI Tier	N	Shock Rate	Balance Change Ratio	Full Drain Rate	Fraud Rate
Low Risk	33,000	0.0000	0.1266	0.1097	0.0001
Medium Risk	33,000	0.0000	-0.8158	0.2543	0.0002
High Risk	34,000	0.1483	-1096.7560	0.3526	0.0039

Table 4: Mean Risk Indicators by FSI Tier - Validation of the Financial Stress Index

The results show a clear and consistent pattern. Shock rate increases sharply from 0% in the Low and Medium Risk tiers to 14.83% in the High Risk tier. Full drain rate also rises steadily across tiers (10.97%, 25.43%, and 35.26%), indicating increasing structural financial stress.

Fraud rate, used only for post-hoc validation, increases significantly from 0.01% in the Low Risk tier to 0.39% in the High Risk tier, representing a substantial escalation in observed fraud incidence. The highly negative balance change ratio in the High Risk tier reflects extreme account depletion events, reinforcing the interpretation of this tier as high-risk.

Overall, the monotonic progression of all key indicators across FSI tiers confirms that the index is a valid and reliable measure of transaction-level financial risk.

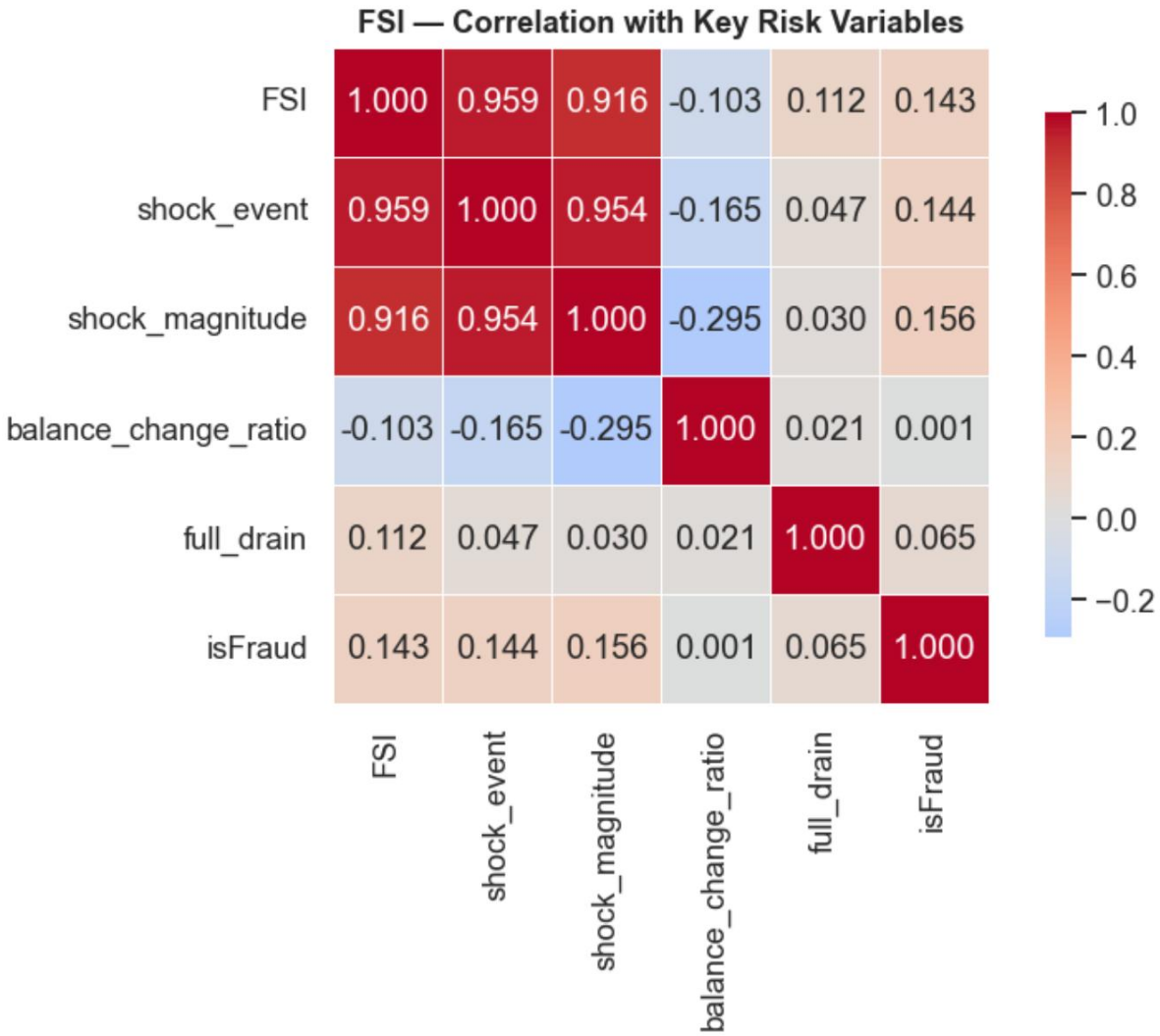

Figure 11: Correlation Heatmap - FSI vs Key Risk Variables

10.3 Linking Results to Real-World Fraud Detection

The pipeline outputs align with known mobile money fraud patterns . The full_drain feature captures account takeover behaviour where entire balances are transferred, while no_dest_credit identifies inconsistencies in destination balance updates.

The High Risk cluster (Cluster 0), with a fraud rate of 2.69% compared to the dataset average of 0.141%, represents a population nearly 19 times more likely to contain fraud. In practice, such transactions would be prioritised for manual review, transaction blocking, or enhanced authentication.

The Financial Stress Index (FSI) enables continuous risk prioritisation rather than binary decisions. Transactions can be ranked by FSI and reviewed in descending order, ensuring efficient allocation of monitoring resources. Its monotonic relationship with fraud confirms that this ranking is statistically meaningful.

11. Discussion

11.1 Strengths of the Approach

The pipeline demonstrates several strengths. The combination of Isolation Forest and DBSCAN captures both global and local anomalies. Their low overlap confirms complementary detection, improving overall recall.

Feature engineering introduces meaningful, domain-driven variables such as full_drain and no_dest_credit, ensuring that the model captures real fraud behaviour rather than relying solely on statistical patterns. Integration of anomaly scores into clustering ensures that segmentation reflects risk characteristics directly. This leads to clear separation of behavioural personas.

The Financial Stress Index provides a transparent and interpretable risk score. Its validation confirms that it captures genuine risk variation rather than arbitrary scaling.

11.2 Limitations

The use of synthetic data limits real-world applicability. Fraud patterns in PaySim may not fully represent real-world complexity.

DBSCAN is applied only to a sample (8%), leaving most transactions without density-based evaluation, which may introduce false negatives.

The Isolation Forest contamination parameter (0.05) is predefined rather than optimised, and its sensitivity is not explored.

The dataset lacks repeated user activity, preventing the use of account-level behavioural features, which are important in real fraud detection systems.

11.3 Challenges Faced

Extreme class imbalance (0.141% fraud) creates a trade-off between precision and recall. The chosen contamination level prioritises recall, which is appropriate for fraud detection.

High dimensionality (15 features) introduces challenges for distance-based clustering, reflected in moderate Silhouette Scores. This is expected in financial datasets where behaviour overlaps.

12. Conclusion

This project demonstrates that financial shocks and behavioural risk profiles can be effectively identified using unsupervised learning without reliance on fraud labels. By leveraging the intrinsic structure of transaction data, the proposed pipeline uncovers latent risk patterns that are not directly observable through traditional supervised approaches.

A key insight is that financial risk is inherently multi-dimensional and cannot be captured through isolated features or single-model frameworks. The integration of engineered structural features, such as balance dynamics and anomaly indicators, with an ensemble anomaly detection strategy enables robust identification of statistically significant deviations in transactional behaviour.

The derived behavioural personas — Stable, Moderate Risk, Shock-Prone, and High Risk — exhibit clear differentiation across all major risk indicators, including shock rate, shock magnitude, full drain rate, fraud rate, and Financial Stress Index (FSI). The High Risk segment is approximately 19 times more likely to contain fraud than the dataset average, demonstrating the practical relevance of the segmentation.

The Financial Stress Index provides a continuous and interpretable risk signal that supports prioritised decision-making. Unlike binary classification systems, it enables graded risk assessment, allowing monitoring resources to be allocated efficiently based on transaction severity.

Importantly, this work establishes that unsupervised learning is not merely an alternative in the absence of labels, but a suitable and effective approach for dynamic financial environments where risk patterns evolve over time.

Overall, the project presents a scalable and interpretable framework for financial risk discovery, with direct applicability to transaction monitoring systems, fraud triage workflows, and risk-based customer profiling in modern digital payment ecosystems.

13. Future Work

Several extensions can further enhance the analytical depth and real-world applicability of this project. A key improvement would be the adoption of scalable density-based clustering methods such as HDBSCAN, which can operate efficiently on the full dataset and eliminate the need for sampling in DBSCAN, thereby improving anomaly coverage. The Isolation Forest model can be further refined by optimising the contamination parameter using Precision–Recall analysis against available validation labels, enabling a more data-driven calibration of anomaly thresholds.

Incorporating account-level temporal features, where multiple transactions per account are available, would significantly strengthen the model's ability to detect evolving behavioural patterns such as gradual account compromise or repeated suspicious activity. The current unsupervised framework can also be extended into a semi-supervised approach, where cluster-derived behavioural personas are used as pseudo-labels to iteratively train and refine a classification model as new labelled data becomes available.

From an operational perspective, implementing the pipeline within a real-time streaming architecture using technologies such as Apache Kafka and Spark Streaming would enable continuous risk scoring and immediate anomaly detection as transactions occur. Additionally, benchmarking the current approach against deep learning techniques, such as autoencoder-based anomaly detection, would provide insights into potential performance improvements in capturing complex non-linear patterns.

Finally, the most critical extension is validation on real-world transaction datasets. Applying the pipeline to actual mobile money or banking data would assess its robustness, generalisability, and practical effectiveness, ensuring that the insights derived are applicable in real financial risk monitoring environments.

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